

2008 Ford Edge SE

2008 ENGINE Engine Cooling - Edge & MKX

2008 ENGINE**Engine Cooling - Edge & MKX****SPECIFICATIONS****MATERIAL**

Item	Specification	Fill Capacity
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1	9.2L (9.7 qt)
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A	-
Premium Cooling System Flush VC-1	ESR-M14P7-A	-

GENERAL SPECIFICATIONS

Item	Specification
Cooling System Pressure Test Specifications	
Cooling System	124 kPa (18 psi)
Radiator Cap Pressure Test Specification	
Pressure release cap	89-124 kPa (13-18 psi)
Thermostat Opening Temperatures	
Thermostat starts to open	86°C (187°F)
Thermostat fully open	96°C (205°F)

TORQUE SPECIFICATIONS

Description	N.m	lb-ft	lb-in
A/C condenser-to-radiator bolts	10	-	89
Block heater	40	30	-
Compressor-to-condenser bracket bolt	7	-	62
Coolant pump bolts ^a	-	-	-
Cooling fan motor and shroud bolts	6	-	53
Degas bottle-to-fender bolt	9	-	80
Engine oil filter ⁽¹⁾	-	-	-
Exhaust clamp	40	30	-

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Oil cooler bracket bolt	6	-	53
Oil pan drain plug	27	20	-
Primary timing chain guide bolts	10	-	89
Primary timing chain tensioner bolts	10	-	89
Thermostat housing bolts	10	-	89
Thermostat housing cover bolts	10	-	89
Variable camshaft timing (VCT) oil control solenoid housing bolts ⁽¹⁾	-	-	-
Y-pipe nuts	40	30	-
(1) Refer to the procedure in this service information.			

DESCRIPTION AND OPERATION

ENGINE COOLING

CAUTION: The engine cooling system is filled with Motorcraft Premium Gold Engine Coolant. Mixing coolant types degrades the corrosion protection of Motorcraft Premium Gold Engine coolant.

NOTE: Stop-leak style pellets/products must not be used as an additive in this engine cooling system. The addition of stop-leak style pellets/products can clog or damage the cooling system, resulting in degraded cooling system performance and/or failure.

The cooling system components are the:

- radiator.
- bypass tube.
- pressure relief cap.
- degas bottle.
- radiator draincock.
- coolant pump.
- engine coolant temperature indicator unit.
- thermostat (cold side).
- variable speed fan motor assembly.
- fan control module.
- engine block coolant weep hole.

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Bezel Diagnostics

On Demand Self Test	Speaker Walk Test	Exit Diagnostics
View DTC's	Multi-disc Test	
Configuration Status	Display Test	
System Info	HardKey Test	
GPS Info	Radio Signal Strength	

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Fig. 1: Locating Engine Block Coolant Weep Hole
Courtesy of FORD MOTOR CO.

The engine is equipped with an engine block coolant weep hole. If the inner coolant pump shaft seal fails, coolant will be diverted out the weep hole in the LH side of the engine block. This prevents any coolant from entering the crankcase.

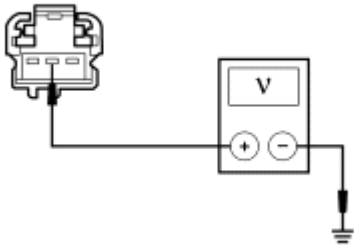
Engine coolant provides freeze protection, boil protection, cooling efficiency and corrosion protection to the engine and cooling components. In order to obtain these protections, the engine coolant must be maintained at the correct concentration and fluid level in the degas bottle.

When adding engine coolant, use a 50/50 mixture of engine coolant and distilled water.

To maintain the integrity of the coolant and the cooling system:

- add Motorcraft Premium Gold Engine Coolant. Do not mix coolant types.
- do not add/mix orange-colored Motorcraft Specialty Orange Engine Coolant or equivalent meeting Ford specification WSS-M97B44-D or green colored Motorcraft Premium Engine Coolant meeting Ford specification ESE-M97B44-A or equivalent. Mixing coolants may degrade the coolant's corrosion protection.
- do not add alcohol, methanol or brine, or any engine coolants mixed with alcohol or methanol antifreeze. These can cause engine damage from overheating or freezing.
- Ford Motor Company does NOT recommend the use of recycled engine coolant in vehicles originally equipped with Motorcraft Premium Gold Engine Coolant since a Ford-approved recycling process is not yet available.

NOTE: In the following illustration, Black arrows indicate hot, white arrows indicate cold.



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Fig. 2: Coolant Flow Diagram, 3.5L
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	8C633	Lower degas bottle hose
2	8005	Radiator
3	8B273	Lower radiator hose
4	8B274	Upper radiator hose
5	6050	Cylinder head
6	6083	Cylinder head gasket
7	8A586	Thermostat housing
8	9N271	Heater inlet tube
9	9K461	Lower intake manifold
10	8501	Coolant pump
11	6010	Engine block
12	8W005	Upper degas bottle hose
13	18C553	Heater hose assembly
14	18B539	Heater core
15	8A080	Degas bottle

DIAGNOSTIC TESTS

ENGINE COOLING

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent

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 ST1720-A	Battery/Antifreeze Tester	014-R1060 or equivalent
 ST1474-A	Pressure Test Kit	014-R1072 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware or equivalent scan tool	-

MATERIAL

Item	Specification
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1

Principles of Operation

Engine coolant flows primarily from the engine to the radiator circuit and back to the coolant pump. From the coolant pump, coolant is sent through the engine block and cylinder heads. A separate circuit from the engine also feeds the heater core with coolant. On most engines, the coolant pump is operated by engine rotation through a pulley which is driven by the accessory drive belt to circulate the coolant. Some engines use a belt driven by a pulley attached to the camshaft, and some engines use a sprocket driven by the timing chain to operate the coolant pump. The coolant thermostat is a control valve actuated by coolant temperature. When the thermostat is closed, coolant flow bypasses the radiator circuit and returns to the coolant pump. When the thermostat is opened, coolant is allowed to flow through the radiator circuit in order to transfer engine generated heat to the outside air.

NOTE: **The vehicle is equipped with either a degas bottle or an expansion tank.**

The degas bottle, if equipped, holds surplus coolant and removes air from the cooling system, which reduces hot spots. It also allows for coolant expansion and system pressurization, replenishes coolant to the cooling system

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and serves as the location for service fill.

The coolant expansion tank, if equipped, holds surplus coolant, allows for coolant expansion and replenishes coolant to the cooling system. It is equipped with a non-pressurized cap which allows coolant to be added to the tank to keep it filled to prescribed levels.

The cooling fan draws air through the radiator to help cool the system coolant as it passes through the radiator.

The thermostat monitor is a function of the PCM and is designed to verify correct thermostat operation. The monitor will be executed once per drive cycle and has a monitor run duration of 300-800 seconds. If a malfunction occurs, DTC P0125 or P0128 is set, and the malfunction indicator lamp will be illuminated.

For vehicle/engine specific information, refer to **ENGINE COOLING** in Description and Operation.

Inspection and Verification

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

CAUTION: The engine cooling system is filled with Motorcraft Premium Gold Engine Coolant. Mixing coolant types degrades the corrosion protection of Motorcraft Premium Gold Engine Coolant. Do not mix coolant types. Failure to follow these instructions may result in engine or cooling system damage.

NOTE: Vehicles equipped with a degas bottle system have the pressure relief cap on the degas bottle and no radiator cap. Vehicles equipped with a coolant expansion tank system have the pressure relief cap on the radiator.

1. Verify the customer concern.
2. Visually check the engine coolant level at the degas bottle or coolant expansion tank when the system is cold.
3. Make sure the pressure relief cap is installed correctly.
4. Record any cooling system DTCs retrieved. Refer to the **PCM DTC CHART** in this service information for DTC descriptions.

NOTE: Take note of any coolant odor or steam coming from cooling system components.

5. If the system coolant is filled correctly and no DTCs associated with fail-safe cooling are retrieved, verify the customer's concern by operating the engine to duplicate the condition.

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NOTE: Refer to the coolant flow diagram in ENGINE COOLING in Description and Operation .

6. Inspect to determine if any of the following mechanical or electrical concerns apply.

VISUAL INSPECTION CHART

Mechanical	Electrical
- Leaks or weeps at: - Hoses - Tubes - Clamp joints - Quick connect couplings (if equipped) - Gaskets - O-rings - Thermostat housing - Radiator - Pressure relief cap - Coolant pump - Heater core (wet floor or coolant odor in vehicle) - Heater control valve - Heated throttle body (TB) or heated TB adapter (if equipped) - Coolant crossover manifold assembly (if equipped) - Oil cooler (if equipped) - Degas bottle (if equipped) - Coolant expansion tank (if equipped)	- Inoperative or damaged: - Electric cooling fan (if equipped) - Electronically actuated fan clutch - Electronically actuated fan clutch controller - Wiring, connectors, relays or modules - Engine coolant temperature (ECT) sensor (if equipped) - Cylinder head temperature (CHT) sensor (if equipped) - Intake air temperature (IAT) sensor (if equipped) - Mass air flow (MAF) sensor - Vehicle speed sensor (VSS)

○ Heated PCV (if equipped) ○ Fuel pressure regulator coolant bowl (if equipped, natural gas engine) ○ Cylinder block core plugs (if equipped) ○ Cylinder head core plugs (if equipped) ○ Block heater (if equipped)	
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VISUAL INSPECTION CHART

Mechanical	Electrical
● Cracked or damaged: ○ Hoses ○ Tubes ○ Hose clamps ○ Heater control valve (if equipped) ○ Thermostat housing ○ Radiator ○ Pressure relief cap ○ Cooling fan ○ Fan clutch (if equipped) ○ Coolant pump ○ Degas bottle (if equipped) ○ Coolant expansion tank (if equipped) ○ Oil cooler (if equipped)	

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- | | |
|---|--|
| <ul style="list-style-type: none">○ Coolant crossover manifold assembly (if equipped)○ Cylinder block core plugs (if equipped)○ Cylinder head core plugs (if equipped)○ Block heater (if equipped)● Restricted airflow through the A/C condenser/radiator● Drive belt loose, worn or installed incorrectly● Broken or weak drive belt tensioner● Excessive white or light gray exhaust smoke (may have burnt coolant odor)● Coolant in engine oil● Engine oil in coolant● Coolant in automatic transmission fluid (if equipped)● Automatic transmission fluid (if equipped) in coolant | |
|---|--|

7. If the inspection reveals an obvious concern that can be readily identified, repair it as necessary. Test the system for normal operation.
8. Inspect the coolant condition in the following sequence:
 1. Inspect the coolant color.
 - If Motorcraft Premium Gold Engine Coolant has a clear or pale yellow color, this indicates higher water content than required. Test the engine coolant freezing point range with the Battery/Antifreeze Tester. The freezing point should be in the range -45°C to -23°C (-50°F to -10°F). If the vehicle is driven in cold climates less than -36°C (-34°F), it may be necessary to increase the coolant concentration to get adequate freeze protection. Recommended coolant concentration is 50/50 ethylene glycol to distilled water.
 - A pale green color indicates incorrect coolant (green in color) may have been added to the

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system. Use of incorrect (green in color) coolant degrades the corrosion protection of Motorcraft Premium Gold Engine Coolant. Flush the system and refill with the correct mixture of distilled water and Motorcraft Premium Gold Engine Coolant.

NOTE: **If Cooling System Stop Leak Pellets are used, darkening of the Motorcraft Premium Gold Engine Coolant from yellow to golden tan will occur.**

- Dark brown could indicate a commercially available stop leak may have been used. Flush the system and refill with the correct mixture of distilled water and Motorcraft Premium Gold Engine Coolant.
 - A light or reddish brown color indicates that rust may be present in the cooling system. Flush the system and refill with the correct mixture of distilled water and Motorcraft Premium Gold Engine Coolant.
 - An iridescent sheen on top of the coolant could indicate a trace of oil is entering the system. For information on engine diagnosis, refer to **ENGINE SYSTEM - GENERAL INFORMATION** article.
 - A milky brown color may indicate that engine oil is entering the cooling system. Pressure test the cooling system. Refer to component tests in this service information. If engine oil is suspected, the cause of the leak may be internal to the engine. Refer to **ENGINE SYSTEM - GENERAL INFORMATION** article.
 - A red, orange or light green colored sheen on top of the coolant may indicate that transmission fluid is entering the cooling system. The cause may be a leaky radiator. Pressure test the cooling system. Refer to **COMPONENT TESTS**.
2. If the engine coolant appearance is acceptable, test the engine coolant freezing point range with the Battery/Antifreeze Tester. The freezing point should be in the range -45°C to -23°C (-50°F to -10°F). If the vehicle is driven in cold climates less than -36°C (-34°F), it may be necessary to increase the coolant concentration to get adequate freeze protection. Recommended coolant concentration is 50/50 ethylene glycol to distilled water.
 - Maximum coolant concentration is 60/40 for cold weather areas.
 - Minimum coolant concentration is 40/60 for warm weather areas.
 3. Adjust coolant range and level if necessary:
 - If coolant is low, add specified coolant mixture only.
 - If the engine coolant tests too weak, remove some of the engine coolant and add undiluted engine coolant until the readings are within acceptable levels.
 - If the engine coolant tests strong, remove some of the engine coolant and add distilled water until the readings are within acceptable levels.
 9. If an obvious cause for an observed or reported concern is found, correct the cause and test the system for normal operation before proceeding to the next step.

NOTE: **Make sure to use the latest scan tool software release.**

10. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

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NOTE: The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.

11. If the scan tool does not communicate with the VCM:
 - check the VCM connection to the vehicle.
 - check the scan tool connection to the VCM.
 - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
12. If the scan tool does not communicate with the vehicle:
 - verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
13. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record continuous memory DTCs.
14. Clear the continuous DTCs and carry out the self-test diagnostics PCM.
15. If the DTCs recovered are related to the concern, go to the PCM DTC Chart. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
16. If no DTCs related to the concern are retrieved, go to **SYMPTOM CHART** table.

PCM DTC CHART

DTC	Description	Action
P0217	Engine Coolant Overtemperature Condition	Go to Pinpoint Test B .
P1285	Cylinder Head Overtemperature Condition	Go to Pinpoint Test B .
P1299	Cylinder Head Overtemperature Protection Active	Go to Pinpoint Test B .
P0125	Insufficient Coolant Temp For Closed Loop Fuel Control	Go to Pinpoint Test c .
P0128	Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)	Go to Pinpoint Test c .
P0480 P0481 P0482	Fan 1, 2 or 3 Control Circuit, Respectively	REFER to the <u>Introduction - Gasoline Engines</u> article.
All Other DTCs	-	REFER to the <u>Introduction - Gasoline Engines</u> article.

Symptom Chart

SYMPTOM CHART

Condition	Possible Sources	Action
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• Loss of coolant	• Coolant hoses or tubes • Hose clamps • Coolant pump O-ring seal or gasket • Thermostat O-ring seal or gasket • Thermostat housing • Radiator • Transmission fluid cooler (in radiator) (if equipped) (may leak internally or externally) • Degas bottle (if equipped) • Coolant expansion tank (if equipped) • Pressure relief cap • Coolant pump leaking from weep hole • Auxiliary coolant pump (if equipped) • Heater core • Heater control valve (if equipped) • Coolant crossover manifold assembly (if equipped) • Engine gaskets (may leak internally or externally) • Oil cooler (if equipped) (may leak internally or externally) • Heated throttle body (TB) or heated TB adapter (if equipped) • Heated PCV (if equipped) • Fuel pressure regulator coolant bowl (if equipped, natural gas engine) • Cylinder block core plugs (if equipped) • Cylinder head core plugs (if equipped) • Block heater (if equipped)	• Go to Pinpoint Test A .
• The engine overheats	• Low coolant level • External engine coolant leak • Airlock in system • Pressure relief cap installation • Restricted airflow through the A/C condenser/radiator	• Go to Pinpoint Test B .

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	• Internal engine coolant leak • Coolant condition/concentration • Accessory drive components • Non-OEM engine enhancement components • Electric cooling fan (if equipped) • Mechanical cooling fan (if equipped) • Mechanical cooling fan clutch (if equipped) • Radiator • Thermostat • Engine coolant temperature (ECT) indicator system (gauge) • ECT sensor (if equipped) • Cylinder head temperature (CHT) sensor (if equipped) • Heater core • Coolant pump • Coolant flow restriction	
• The engine does not reach normal operating temperature	• Low coolant level • Thermostat • Mechanical fan clutch always engaged • Electric cooling fan always on • Engine coolant temperature indicator system (gauge) • ECT sensor (if equipped) • CHT sensor (if equipped)	• Go to Pinpoint Test C .
• The block heater does not operate correctly	• Block heater power cable • Block heater	• CHECK continuity in all 3 power cable circuits. If any circuit measures greater than 5 ohms, INSTALL a new power cable. • CHECK the resistance of the block heater. If the resistance does not measure between 12.5 and 17.0 ohms, INSTALL a new block

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• The electric cooling fan (s) or electronically controlled fan clutch is inoperative in one or more speeds or does not operate correctly (if equipped)	• Wiring • Relays • Fuses • Fan control module • Cooling fan motor(s) • Cooling fan resistor(s) • Electric fan clutch	heater. • REFER to the <u>Introduction - Gasoline Engines</u> article.
• The electric cooling fan (s) stay(s) on all the time (if equipped)	• Wiring • Relays	• REFER to the <u>Introduction - Gasoline Engines</u> article.
• Noisy electric cooling fan operation (if equipped) • Noisy mechanical fan operation (if equipped)	• Foreign material contamination • Fan motor • Fan blade detached from fan motor • Fan clutch	• REMOVE the foreign material from the cooling fan and shroud. TEST the system for normal operation. If still noisy, INSTALL a new cooling fan assembly. • INSTALL a new cooling fan assembly. • CARRY OUT the Fan Clutch Test - Minimum Speed Requirement component test.

Pinpoint Tests

Pinpoint Test A: Loss of Coolant

Normal Operation

The engine cooling system is a closed system that provides for coolant expansion and contraction and also changes in pressure as coolant warms and cools with engine operation. Various gaskets, seals, hoses and clamps are used to contain coolant within the cooling system and keep other fluids and contaminants from entering the cooling system.

Coolant loss can be attributed to either external or internal leaks anywhere within the cooling system.

For vehicle/engine specific information, refer to **ENGINE COOLING** in Description and Operation.

This pinpoint test is intended to diagnose the following:

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- Coolant hoses or tubes
- Hose clamps
- Thermostat O-ring seal or gasket
- Coolant pump O-ring seal or gasket
- Thermostat housing
- Heater control valve (if equipped)
- Radiator
- Transmission fluid cooler (in radiator) (if equipped)
- Pressure relief cap
- Coolant pump leaking from weep hole
- Auxiliary coolant pump (if equipped)
- Heater core
- Heated throttle body (TB) or heated TB adapter (if equipped)
- Coolant crossover manifold assembly (if equipped)
- Engine gaskets
- Oil cooler (if equipped)
- Degas bottle (if equipped)
- Coolant expansion tank (if equipped)
- Heated PCV (if equipped)
- Fuel pressure regulator coolant bowl (if equipped, natural gas engine)
- Cylinder block core plugs (if equipped)
- Cylinder head core plugs (if equipped)
- Block heater (if equipped)
- Engine coolant temperature (ECT) sensor (if equipped)

PINPOINT TEST A: LOSS OF COOLANT

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

A1 CARRY OUT INSPECTION AND VERIFICATION

- Carry out the Inspection and Verification procedure in this service information.
- **Were any concerns found?**
YES : REPAIR as needed. TEST the system for normal operation.
NO : Go to A2 .

A2 CHECK THE ENGINE COOLANT LEVEL

NOTE: Allow the engine to cool before checking the engine coolant level.

- Key in OFF position.
- Visually inspect the engine coolant level at the degas bottle or the coolant expansion tank.
- **Is the engine coolant level within specifications?**

YES : Go to A3.

NO : ADJUST the engine coolant level as necessary. Go to A3.

A3 PRESSURE TEST THE ENGINE COOLING SYSTEM

- Pressure test the engine cooling system. Refer to Cooling System Pressure Test in **COMPONENT TESTS** in this service information.
- **Does the engine cooling system leak?**

YES : REPAIR or INSTALL new components. TEST the system for normal operation.

NO : If the vehicle is equipped with a coolant expansion tank, go to A4. If the vehicle is equipped with a degas bottle, go to A5.

A4 CHECK THE PRESSURE RELIEF CAP

- Carry out the pressure relief cap pressure test. Refer to Pressure Relief Cap Pressure Test in **COMPONENT TESTS** in this service information.
- **Is the pressure relief cap OK?**

YES : Go to A5.

NO : INSTALL a new pressure relief cap. TEST the system for normal operation.

A5 CHECK THE ENGINE COOLANT FOR AN INTERNAL LEAK

- Inspect the engine coolant in the degas bottle or coolant expansion tank for signs of engine oil or transmission fluid.
- **Is engine oil or transmission fluid evident in the coolant?**

YES : If engine oil is evident, go to **ENGINE SYSTEM - GENERAL INFORMATION** article for engine diagnosis. If transmission fluid is evident, INSTALL a new radiator. REPAIR the transmission as necessary. Refer to the appropriate Automatic Transmission article for the procedure. TEST the system for normal operation.

NO : Go to A6.

A6 CHECK THE ENGINE OIL AND TRANSMISSION FLUID FOR COOLANT

- Remove the oil level indicators from the engine and the transmission.
- **Is coolant evident in the oil or transmission fluid?**

YES : If coolant is in the engine oil, go to **ENGINE SYSTEM - GENERAL INFORMATION** article. If coolant is in the transmission fluid, INSTALL a new radiator. REPAIR the transmission as necessary. Refer to the appropriate Automatic Transmission article for the procedure.

NO : Go to A7.

A7 CHECK THE COOLING SYSTEM FOR COMBUSTION GASES

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NOTE: Use U-View® Combustion Leak Tester part number 560000 or equivalent.

- Using a cooling system combustion gas leak tester, following the instructions supplied with the tester, check the coolant for combustion gases.
- **Are combustion gases present?**
YES : Go to **ENGINE SYSTEM - GENERAL INFORMATION** article for engine diagnosis.
NO : The cooling system is operational.

Pinpoint Test B: The Engine Overheats

Normal Operation

The engine cooling system functions to maintain engine temperatures during operation. Correct coolant flow through the engine, radiator and remainder of cooling system passages and components is essential to maintaining a correct engine temperature.

Engine coolant flows primarily from the engine to the radiator circuit and back to the coolant pump. From the coolant pump, coolant is sent through the engine block and cylinder heads. A separate circuit from the engine also feeds the heater core with coolant. On most engines, the coolant pump is operated by engine rotation through a pulley which is driven by the accessory drive belt to circulate the coolant. Some engines use a belt driven by a pulley attached to the camshaft, and some engines use a sprocket driven by the timing chain to operate the coolant pump. The coolant thermostat is a control valve actuated by coolant temperature. When the thermostat is closed, coolant flow bypasses the radiator circuit and returns to the coolant pump. When the thermostat is opened, coolant is allowed to flow through the radiator circuit in order to transfer engine generated heat to the outside air.

Engine overheating generally occurs when there is a disruption in the ability to control either coolant flow at the correct rate, the inability to transfer heat from the engine through the coolant (including low coolant), or an inability to transfer engine generated heat to the outside air through the radiator.

For vehicle/engine specific information, refer to **ENGINE COOLING** in Description and Operation.

- DTC P0217 Engine Coolant Overtemperature Condition
- DTC P1285 Cylinder Head Overtemperature Condition
- DTC P1299 Cylinder Head Overtemperature Protection Active

This pinpoint test is intended to diagnose the following:

- Low coolant level
- External engine coolant leak
- Airlock in system
- Pressure relief cap installation
- Restricted airflow through the A/C condenser/radiator

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- Internal engine coolant leak
- Coolant condition/concentration
- Accessory drive components
- Non-OEM engine enhancement components
- Electric cooling fan (if equipped)
- Mechanical cooling fan blade (if equipped)
- Mechanical cooling fan clutch (if equipped)
- Engine coolant temperature (ECT) indicator system (gauge)
- ECT sensor (if equipped)
- Cylinder head temperature (CHT) sensor (if equipped)
- Heater core
- Coolant pump
- Coolant flow restriction

PINPOINT TEST B: THE ENGINE OVERHEATS

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

B1 CARRY OUT INSPECTION AND VERIFICATION

- Carry out the Inspection and Verification procedure in this service information.
- **Were any concerns found?**

YES : REPAIR as needed. TEST the system for normal operation.

NO : Go to B2 .

B2 CHECK FOR DTCs

NOTE: Refer to the INTRODUCTION - GASOLINE ENGINES article for correct scan tool hook-up procedure.

- Check for DTC P0217, P1285 or P1299.
- **Is DTC P0217, P1285 or P1299 present?**

YES : Go to B3.

NO : Actual engine overheating has not been verified. CHECK the ECT gauge operation. REFER to INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES article. If any other DTCs are retrieved, REFER to the INTRODUCTION - GASOLINE ENGINES article.

B3 CHECK FOR AN AIRFLOW OBSTRUCTION

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- Check the radiator or A/C condenser for an obstruction such as leaves or cardboard.
- **Is an obstruction present?**

YES : REMOVE the obstruction. TEST the system for normal operation.

NO : Go to B4.

B4 CHECK THE ENGINE COOLANT LEVEL

NOTE: **Allow the engine to cool before checking the coolant level.**

- Key in OFF position.
- Visually check the engine coolant level in the degas bottle or coolant expansion tank.
- **Is the engine coolant level within specification?**

YES : Go to B5.

NO : ADJUST the engine coolant level as necessary. Go to B5.

B5 PRESSURE TEST THE ENGINE COOLING SYSTEM

- Pressure test the engine cooling system. Refer to Cooling System Pressure Test in **COMPONENT TESTS** in this service information.
- **Does the engine cooling system leak?**

YES : REPAIR or INSTALL new components. TEST the system for normal operation.

NO : Vehicles equipped with an expansion tank, go to B6. Vehicles equipped with a degas bottle, go to B7.

B6 CHECK THE PRESSURE RELIEF CAP

- Carry out the pressure relief cap pressure test. Refer to Pressure Relief Cap Pressure Test in **COMPONENT TESTS** in this service information.
- **Is the pressure relief cap OK?**

YES : Go to B7.

NO : INSTALL a new pressure relief cap. TEST the system for normal operation.

B7 CHECK THE ENGINE COOLANT FOR AN INTERNAL LEAK

- Inspect the engine coolant in the degas bottle or coolant expansion tank for signs of engine oil or transmission fluid.
- **Is engine oil or transmission fluid evident in the coolant?**

YES : If engine oil is evident, go to **ENGINE SYSTEM - GENERAL INFORMATION** article for engine diagnosis. If transmission fluid is evident, INSTALL a new radiator. REPAIR the transmission as necessary. Refer to the appropriate Automatic Transmission article for the procedure. TEST the system for normal operation.

NO : Go to B8.

B8 CHECK THE ENGINE OIL AND TRANSMISSION FLUID FOR COOLANT

- Remove the oil level indicators from the engine and the transmission.
- **Is coolant evident in the oil or transmission fluid?**

YES : If coolant is in the engine oil, go to **ENGINE SYSTEM - GENERAL INFORMATION**

article. If coolant is in the transmission fluid, INSTALL a new radiator. REPAIR the transmission as necessary. Refer to the appropriate Automatic Transmission article for the procedure.

NO : Go to B9.

B9 CHECK THE COOLING SYSTEM FOR COMBUSTION GASES

NOTE: Use U-View® Combustion Leak Tester part number 560000 or equivalent.

- Using a cooling system combustion gas leak tester, following the instructions supplied with the tester, check the coolant for combustion gases.
- **Are combustion gases present?**

YES : Go to ENGINE SYSTEM - GENERAL INFORMATION article for engine diagnosis.

NO : Go to B10.

B10 CHECK COOLANT CONDITION

- Check the coolant for dirt, rust or contamination and check the coolant concentration.
- **Is the coolant condition OK?**

YES : Vehicles equipped with an electric cooling fan, go to B11 . Vehicles equipped with only a mechanical cooling fan, go to B12 .

NO : FLUSH the engine cooling system. REFER to COOLING SYSTEM FLUSHING. TEST the system for normal operation.

B11 CHECK THE ELECTRIC COOLING FAN OPERATION

- Start the engine.
- Place the climate control function selector in the MAX A/C position and the blower motor switch in the HI position.
- **Did the electric cooling fan operate?**

YES : If the vehicle is also equipped with a mechanical cooling fan, go to B12. Otherwise, go to B13.

NO : DIAGNOSE the electric cooling fan operation. REFER to INTRODUCTION - GASOLINE ENGINES article.

B12 CHECK THE MECHANICAL COOLING FAN OPERATION

- Key in OFF position.
- If the vehicle is equipped with a viscous fan clutch, carry out the Fan Clutch Tests in COMPONENT TESTS. If the vehicle is equipped with an electronically actuated fan clutch, refer to the INTRODUCTION - GASOLINE ENGINES article.
- **Is the cooling fan operation OK?**

YES : Go to B13.

NO : INSTALL a new fan clutch. TEST the system for normal operation.

B13 CHECK THE COOLANT PUMP OPERATION

- Start the engine.
- Allow the engine to run for 10 minutes (hot side thermostat equipped vehicles) or 30 minutes (cold

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side thermostat equipped vehicles). Place the climate control function selector in the MAX HEAT position. Feel the heater outlet hose.

- **Is the heater outlet hose hot?**

YES : Go to B14.

NO : INSTALL a new coolant pump. TEST the system for normal operation.

B14 CHECK THE THERMOSTAT OPERATION

- Start the engine.
- Allow the engine to run for 10 minutes (hot side thermostat equipped vehicles) or 30 minutes (cold side thermostat equipped vehicles). Feel the upper radiator hose (hot side thermostat equipped vehicles) or the lower radiator hose (cold side thermostat equipped vehicles).
- **Is the upper (hot side thermostat equipped vehicles) or lower (cold side equipped vehicles) radiator hose hot?**

YES : CHECK the engine coolant temperature gauge operation. REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article.

NO : Go to B15.

B15 VISUALLY INSPECT THE THERMOSTAT

- Carry out the Thermostat Visual Inspection in **COMPONENT TESTS**.
- **Is the thermostat damaged?**

YES : INSTALL a new thermostat. TEST the system for normal operation.

NO : INSTALL a new thermostat. TEST the system for normal operation. If the engine still overheats, INSTALL a new radiator. TEST the system for normal operation.

Pinpoint Test C: The Engine Does Not Reach Normal Operating Temperature

Normal Operation

The engine cooling system functions to maintain engine temperatures during operation. Correct coolant flow through the engine, radiator and remainder of cooling system passages and components is essential to maintaining a correct engine temperature.

Engine coolant flows primarily from the engine to the radiator circuit and back to the coolant pump. From the coolant pump, coolant is sent through the engine block and cylinder heads. A separate circuit from the engine also feeds the heater core with coolant. On most engines, the coolant pump is operated by engine rotation through a pulley which is driven by the accessory drive belt to circulate the coolant. Some engines use a belt driven by a pulley attached to the camshaft, and some engines use a sprocket driven by the timing chain to operate the coolant pump. The coolant thermostat is a control valve actuated by coolant temperature. When the thermostat is closed, coolant flow bypasses the radiator circuit and returns to the coolant pump. When the thermostat is opened, coolant is allowed to flow through the radiator circuit in order to transfer engine generated heat to the outside air.

Concerns of engine inability to reach normal operating temperature typically occur when the rate of coolant flow through some coolant circuits (radiator, heater core) is more than expected given the conditions, or when the cooling fans operate all the time (electric fans) or the fan clutch is always engaged (engine driven fans). Heat is not allowed to build in the engine because a heat exchanger is removing too much heat including the

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radiator, heater core and oil cooler. In addition, perceived concerns that the engine does not reach normal operating temperature can be related to a low coolant level or trapped air which does not allow for hot coolant to be available at the heater core, an inoperative climate control system, or for concerns perceived or related to an incorrect engine temperature gauge indication.

For vehicle/engine specific information, refer to **ENGINE COOLING** in Description and Operation.

- DTC P0125 Insufficient Coolant Temp for Closed Loop Fuel Control
- DTC P0128 Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature)

This pinpoint test is intended to diagnose the following:

- Low coolant level
- Thermostat
- Engine coolant temperature (ECT) indicator system (gauge)
- Engine cooling fan

PINPOINT TEST C: THE ENGINE DOES NOT REACH NORMAL OPERATING TEMPERATURE

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

C1 CARRY OUT INSPECTION AND VERIFICATION

- Carry out the Inspection and Verification procedure in this service information.
- **Were any concerns found?**

YES : REPAIR as needed. TEST the system for normal operation.

NO : Go to C2.

C2 CHECK FOR DTCs P0125 or P0128

NOTE: Refer to the **INTRODUCTION - GASOLINE ENGINES** article for correct scan tool hook-up procedure.

- Check for DTC P0125 or P0128.
- **Is DTC P0125 or P0128 present?**

YES : If the vehicle is equipped with an electric cooling fan, go to C3. If the vehicle is equipped with only a mechanical cooling fan, go to C4.

NO : The cooling system is operational. If an inoperative engine coolant temperature gauge is suspected, CHECK the engine coolant temperature gauge operation. REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article. If an inoperative

climate control system is suspected, CHECK the climate control system operation. REFER to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article. If any other DTCs are retrieved, REFER to the **INTRODUCTION - GASOLINE ENGINES** article.

C3 CHECK THE ELECTRIC COOLING FAN OPERATION

- Allow the engine to cool.
- Make sure the A/C switch is OFF (if equipped).
- Start the engine.
- Check the electric cooling fan.
- **Is the electric cooling fan on all the time?**

YES : DIAGNOSE the electric cooling fan operation. REFER to the **INTRODUCTION - GASOLINE ENGINES** article.

NO : If the vehicle is also equipped with a mechanical cooling fan, go to C4. Otherwise, go to C5.

C4 CHECK THE MECHANICAL COOLING FAN OPERATION

- Carry out the Cooling Fan Clutch Component Test. For a mechanical cooling fan clutch, refer to **COMPONENT TESTS** in this service information. For an electronic cooling fan clutch, refer to the **INTRODUCTION - GASOLINE ENGINES** article.
- **Is the cooling fan clutch OK?**

YES : Go to C5.

NO : INSTALL a new cooling fan clutch. TEST the system for normal operation.

C5 CHECK THE COOLANT LEVEL

NOTE: **Allow the engine to cool before checking the coolant level.**

- Visually check the engine coolant level in the degas bottle or coolant expansion tank.
- **Is the engine coolant level within specification?**

YES : INSTALL a new thermostat. TEST the system for normal operation.

NO : Go to **Pinpoint Test A** to diagnose a coolant leak.

Component Tests**Pressure Test - Degas Bottle Systems**

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

NOTE: Vehicles equipped with a degas bottle system have the pressure relief cap on the degas bottle and no radiator cap. Vehicles equipped with a coolant

expansion tank system have the pressure relief cap on the radiator.

1. Turn the engine OFF.
2. Check the engine coolant level. Adjust the coolant level as necessary.
3. Attach the Radiator/Heater Core Pressure Tester to the degas bottle nipple and overflow hose. Install a pressure test pump to the quick connect fitting of the test adapter.

CAUTION: Do not pressurize the cooling system beyond the maximum pressure listed in SPECIFICATIONS in this service information or cooling system components may be damaged.

NOTE: If the plunger of the pressure tester is depressed too fast, an erroneous pressure reading will result.

4. Slowly depress the plunger of the pressure test pump until the pressure gauge reading stops increasing and note the highest pressure reading obtained. If the pressure reading exceeds the maximum cap pressure listed in the SPECIFICATIONS , install a new pressure relief cap.
5. If the system does not hold pressure, remove the pressure relief cap and wash in clean water to dislodge all the foreign material from the gasket. Check the sealing surface in the filler neck of the degas bottle for nicks or cuts. Install the pressure relief cap.
6. Pressurize the engine cooling system as described in Step 4 above. Observe the gauge reading for approximately 2 minutes. Pressure should not drop during this time. If the pressure drops within this time, inspect for leaks and repair as necessary.
7. If no leaks are found and the pressure drops. the pressure relief cap may be leaking. Install a new pressure relief cap and retest the system.
8. If no leaks are found after a new pressure relief cap is installed, and the pressure drops, the leak may be internal to the radiator transmission cooler (if equipped). Inspect the coolant for transmission fluid and the transmission fluid for coolant. Repair as necessary.
9. If there is no contamination of the coolant or transmission fluid, the leak may be internal to the engine. Inspect the coolant for engine oil and the engine oil for coolant. Refer to ENGINE SYSTEM - GENERAL INFORMATION article to diagnose the engine.
10. Release the system pressure by loosening the pressure relief cap. Check the coolant level and adjust as necessary.

Pressure Test - Coolant Expansion Tank Systems

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

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NOTE: Vehicles equipped with a degas bottle system have the pressure relief cap on the degas bottle and no radiator cap. Vehicles equipped with a coolant expansion tank system have the pressure relief cap on the radiator.

1. Turn the engine OFF.
2. Remove the pressure relief cap. Top off the radiator as needed. Fit the pressure tester to the radiator fill neck.

CAUTION: Do not pressurize the cooling system beyond 138 kPa (20 psi) or cooling system components may be damaged.

NOTE: If the plunger of the pressure tester is depressed too fast, an incorrect pressure reading will result.

3. Pump the cooling system to a maximum of 138 kPa (20 psi) and hold for 2 minutes. If the pressure drops within this time, inspect for leaks and repair as necessary.
4. If no leaks are found and the pressure drops, the leak may be internal to the radiator transmission cooler (if equipped). Inspect the coolant for transmission fluid and the transmission fluid for coolant. Repair as necessary.
5. If there is no contamination of the coolant or transmission fluid, the leak may be internal to the engine. Inspect the coolant for engine oil and the engine oil for coolant. Refer to **ENGINE SYSTEM - GENERAL INFORMATION** article to diagnose the engine.

Cap

1. Inspect the pressure relief cap and seals for damage or deterioration. Install a new pressure relief cap if necessary.
2. Fit the pressure relief cap to the Radiator/Heater Core Pressure Tester Kit using the correct adapter.

NOTE: If the plunger of the pressure tester is depressed too fast, an incorrect pressure reading will result.

3. Slowly pump the pressure tester until the pressure gauge stops increasing and note the highest pressure reading. Release the pressure and repeat the test. Install a new pressure relief cap if the pressure is not within specification. Refer to the **GENERAL SPECIFICATIONS** table in this service information.

Thermostat

A new thermostat should be installed only after the following tests and checks have been carried out:

- Pinpoint Test A, B or C
- Thermostat Visual Inspection

Thermostat Visual Inspection

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1. Remove the thermostat.
2. Examine the thermostat for signs of damage including:
 - Valve not fully seated (light visible through the valve)
 - Foreign material lodged in the main valve
 - Bent or broken frame or flange
 - Bent or broken spring
 - Bent or broken valve or valve stem
 - Wax leaking from wax reservoir or a bulge in the reservoir
 - Any other damage or distortion

NOTE: If no damage is found during the inspection, do not attempt to open the thermostat using hot water or other heat sources. This method is not an accurate means to test the function of the thermostat and may damage the thermostat.

3. If damage is found during the inspection, remove any foreign material or broken pieces and install a new thermostat.
4. If no damage is found during the inspection, continue troubleshooting the system concern. Go to the Symptom Chart for further instructions.

Radiator Leak Test, Removed From Vehicle

CAUTION: Never leak test an aluminum radiator in the same water that copper/brass radiators are tested in. Flux and caustic cleaners may be present in the cleaning tank and they will damage aluminum radiators.

NOTE: Always install plugs in the oil cooler fittings before leak testing or cleaning any radiator.

NOTE: Clean the radiator before leak testing to avoid contamination of tank.

1. Leak test the radiator in clean water with 138 kPa (20 psi) air pressure.

Fan Clutch Test

1. Turn the engine OFF and wait until the fan comes to a complete stop.
2. Spin the fan blade by hand. A light resistance should be felt. If there is no resistance or very high resistance, the minimum and maximum fan speeds must be checked. For an electronically actuated fan clutch, refer to the **INTRODUCTION - GASOLINE ENGINES** article. For a viscous fan clutch, carry out the following:

Fan Clutch Test - Minimum Speed Requirement

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1. Use a suitable marker to mark the coolant pump or fan pulley and one of the fan blades.
2. Make sure the A/C is OFF, if equipped.

WARNING: Do not operate the engine with the hood open until the fan blade has been examined for possible cracks and separation. A damaged fan can separate during operation. Failure to follow this instruction may result in serious personal injury.

3. Start the engine and run it at approximately 2,000 RPM for 5 minutes or until there is a noticeable reduction in fan noise to allow the fan clutch to go into disengaged mode.
4. Aim a laser photo tachometer at the coolant pump or fan pulley. Run the engine to achieve 3,000 RPM at the coolant pump or fan pulley.
5. With the coolant pump or fan pulley at 3,000 RPM, aim the laser photo tachometer at the fan blade. Monitor and record fan speed.
6. The fan blade speed must be less than the specified RPM at 3,000 RPM coolant pump or fan pulley RPM. Refer to **SPECIFICATIONS** table in this service information for correct fan speed.
7. Turn the engine off.
8. If the fan blade speed was greater than specified, install a new fan clutch.

Fan Clutch Test - Maximum Speed Requirement

1. Use a suitable marker to mark the coolant pump or fan pulley and one of the fan blades.
2. Block off areas on each side of the radiator in the engine compartment, the front of the radiator grille and the bumper. Close the hood. This will raise the temperature of the air striking the fan clutch and should cause the fan blade to operate at maximum speed.
3. Place the climate control function selector switch in the MAX A/C position and the blower motor switch in the HI position, if equipped.

WARNING: Do not operate the engine with the hood open until the fan blade has been examined for possible cracks and separation. A damaged fan can separate during operation. Failure to follow this instruction may result in serious personal injury.

NOTE: Do not open the hood to check the coolant pump or fan pulley temperature. This will lower the temperature of the air reaching the fan clutch and void the test. Aim the infrared thermometer through the wheel well or from under the vehicle.

4. Start the engine and run it at approximately 2,000 RPM until normal operating temperature has been achieved. Using an infrared thermometer, monitor the coolant pump or fan pulley. Run the engine until the coolant pump or fan pulley is at least 96°C (205°F).

NOTE: Do not open the hood to check the coolant pump or fan pulley or fan blade

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speed. This will lower the temperature of the air reaching the fan clutch and void the test. Aim the laser photo tachometer through the wheel well or from under the vehicle.

5. Aim a laser photo tachometer at the coolant pump or fan pulley. Run the engine to achieve 3,000 RPM at the coolant pump or fan pulley.
6. With the coolant pump or fan pulley at 3,000 RPM, aim the laser photo tachometer at the fan blade. Monitor and record fan blade speed.
7. The fan blade speed must be greater than the specified RPM at 3,000 RPM coolant pump or fan pulley RPM. Refer to **SPECIFICATIONS** in this service information for correct fan speed.
8. Open the hood and allow the engine to idle momentarily to lower engine temperature. Turn the engine off and remove the blocks from the radiator, grille and bumper.
9. If the fan blade speed is less than specified, install a new fan clutch.

GENERAL PROCEDURES

COOLING SYSTEM DRAINING, FILLING AND BLEEDING

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST2B18-A	RADKITPLUS	078-00497

MATERIAL

Item	Specification
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1

Draining

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

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CAUTION: The coolant must be recovered in a suitable, clean container for reuse. If the coolant is contaminated, it must be recycled or disposed of correctly. Using contaminated coolant may result in damage to the engine or cooling system components.

CAUTION: The engine cooling system is filled with Motorcraft Premium Gold Engine Coolant. Mixing coolant types degrades the corrosion protection of Motorcraft Premium Gold Engine Coolant.

NOTE: Stop-leak style pellets/products must not be used as an additive in this engine cooling system. The addition of stop-leak style pellets/products can clog or damage the cooling system resulting in degraded cooling system performance and/or failure.

NOTE: Less than 80% of coolant capacity can be recovered with the engine in the vehicle. Dirty, rusty or contaminated coolant requires replacement.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Release the pressure in the cooling system by slowly turning the pressure relief cap 1/2 turn counterclockwise. When the pressure is released, remove the pressure relief cap.
3. Place a suitable container below the radiator draincock.
 - Open the draincock and allow to drain.
 - Close the draincock after draining.
4. When the coolant is drained, make sure drain plugs are installed and tight.

Filling and Bleeding with RADKITPLUS

1. Using the special tool, install the RADKITPLUS and follow the manufacturer's instructions to fill and bleed the cooling system.

Filling and Bleeding without RADKITPLUS

CAUTION: Engine coolant provides freeze protection, boil protection, cooling efficiency and corrosion protection to the engine and cooling components. In order to obtain these protections, the engine coolant must be maintained at the correct concentration and fluid level in the degas bottle.

When adding engine coolant, use a 50/50 mixture of engine coolant and distilled water.

To maintain the integrity of the coolant and the cooling system:

- Vehicle cooling systems are filled with Motorcraft Premium Gold

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Engine Coolant. Always fill the cooling system with the same coolant that is present in the system. Do not mix coolant types.

- Do not add/mix orange-colored Motorcraft Specialty Orange Engine Coolant or equivalent, meeting Ford specification WSS-M97B44-D or green colored Premium Engine Coolant. Mixing coolants may degrade the coolant's corrosion protection.
- Do not add alcohol, methanol or brine, or any engine coolants mixed with alcohol or methanol antifreeze. These can cause engine damage from overheating or freezing.
- Ford Motor Company does NOT recommend the use of recycled engine coolant in vehicles originally equipped with Motorcraft Premium Gold Engine Coolant since a Ford-approved recycling process is not yet available.
- Used engine coolant should be disposed of in an appropriate manner. Follow your community's regulations and standards for recycling and disposing of automotive fluids.

NOTE: Stop-leak style pellets/products must not be used as an additive in this engine cooling system. The addition of stop-leak style pellets/products can clog or damage the cooling system resulting in degraded cooling system performance and/or failure.

1. Open the degas bottle cap and fill the degas bottle to the MAX fill line.
2. Close the degas bottle cap.

CAUTION: If the engine overheats or the fluid level drops below the minimum fill line, shut off the engine and add fluid to the degas bottle maximum fill line once the engine cools. Failure to follow these instructions may result in damage to the engine.

3. Start the engine and let idle for 10 minutes or until the engine reaches normal operating temperature.
4. Repeat Step 3 if necessary.
5. Start the engine and turn the heater to the MAX position.

CAUTION: If the engine overheats or the fluid level drops below the minimum fill line, shut off the engine and add fluid to the degas bottle maximum fill line once the engine cools. Failure to follow these instructions may result in damage to the engine.

6. Run the engine at 2,500 RPM for 10 minutes.
7. Repeat Step 6 if necessary.

WARNING: Always allow the engine to cool before opening the cooling system.

Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

8. Check the engine coolant level in degas bottle and fill as necessary.

COOLING SYSTEM FLUSHING

MATERIAL

Item	Specification
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1
Premium Cooling System Flush VC-1	ESR-M14P7-A

1. To remove rust, sludge and other foreign material from the cooling system, use cooling system flush that is safe for use with aluminum radiators, For additional information, refer to **SPECIFICATIONS**. This cleaning restores cooling system efficiency and helps prevent overheating. A pulsating or reversed direction of flushing water will loosen sediment more quickly than a steady flow in the normal coolant flow direction. In severe cases where cleaning solvents will not clean the cooling system efficiently, it will be necessary to use the pressure flushing method using cooling system flusher. Dispose of old coolant and flushing water contaminated with antifreeze and cleaning chemicals in accordance with local, state or federal laws.
2. Remove the radiator. For additional information, refer to **RADIATOR**.

CAUTION: Radiator internal pressure must not exceed 138 kPa (20 psi). Damage to the radiator can result.

3. Backflush the radiator with the radiator in an upside-down position with a high-pressure hose in the lower hose location and backflush.
4. Remove the thermostat. For additional information, refer to **THERMOSTAT**.
5. Backflush the engine. Position the high-pressure water hose into the engine through the engine return and backflush the engine.

REMOVAL AND INSTALLATION

BLOCK HEATER

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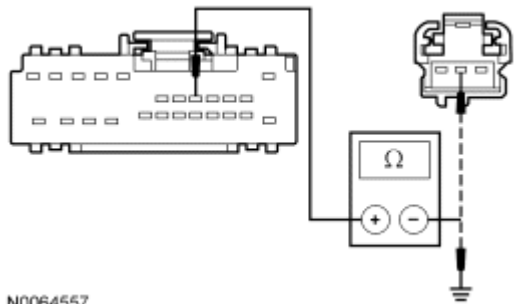


Fig. 3: Exploded View Of Block Heater With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	030104	Block heater cover
2	6B018	Block heater electrical connector
3	6A051	Block heater

REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Drain the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.
3. Remove the RH catalytic converter. For additional information, refer to **EXHAUST SYSTEM** article.
4. Remove the block heater cover.

NOTE: Make sure that the block heater wiring is routed and secured away from rotating or hot components, or damage to the wiring may occur.

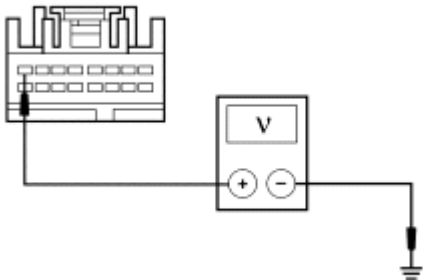
5. Disconnect the block heater electrical connector.
6. Remove the block heater.
 - To install, tighten to 40 N.m (30 lb-ft).
7. To install, reverse the removal procedure.
8. Fill and bleed the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.

THERMOSTAT

MATERIAL

Item	Specification
Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent	WSS-M97B51-A1

(yellow color)	
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N0056311

Fig. 4: Exploded View Of Thermostat With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W500014	Thermostat housing cover bolt (2 required)
2	8594	Thermostat housing cover
3	-	O-ring seal
4	8575	Thermostat

REMOVAL AND INSTALLATION

1. Drain the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.
2. Remove the air cleaner assembly and outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
3. Remove the 2 bolts and position aside the thermostat housing cover.
 - To install, tighten to 10 N.m (89 lb-in).
4. Remove the O-ring seal and thermostat.
 - Clean and inspect the O-ring seal. Install a new seal if necessary.

NOTE: **Lubricate the thermostat O-ring seal with clean engine coolant.**

5. To install, reverse the removal procedure.
6. Fill and bleed the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.

THERMOSTAT HOUSING

MATERIAL

Item	Specification

Motorcraft Premium Gold Engine Coolant with Bittering Agent (US only) VC-7-B (US); CVC-7-A (Canada); or equivalent (yellow color)	WSS-M97B51-A1
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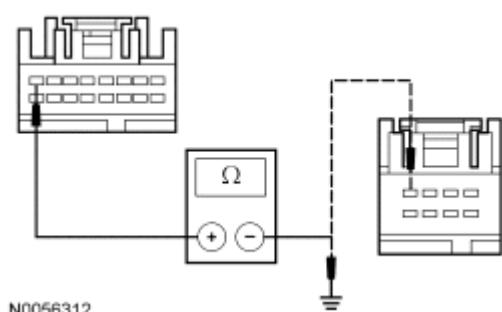


Fig. 5: Identifying Thermostat Housing With Torque Specifications
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	8W005	Vent hose
2	8B273	Lower radiator hose
3	8B274	Upper radiator hose
4	18K580	Heater core outlet hose
5	18K579	Heater core inlet hose
6	W503279	Thermostat housing bolt (3 required)
7	8565	O-ring seal
8	8A571	O-ring seal
9	8A587	Thermostat housing

REMOVAL AND INSTALLATION

1. Drain the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING.**
2. Remove the air cleaner outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
3. Disconnect the 5 coolant hoses from the thermostat housing and position them aside.
4. Remove the 3 thermostat housing bolts.
 - To install, tighten to 10 N.m (89 lb-in).

NOTE: **Do not pull the other end of the coolant tube out of the engine block when separating the thermostat housing.**

5. Separate the thermostat housing from the coolant tube and remove the thermostat housing.
 - Remove and discard the O-ring seals.

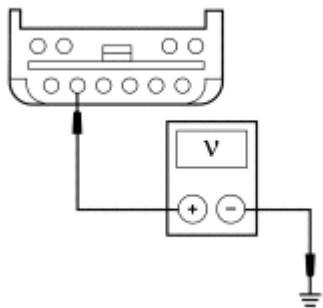
6. To install, reverse the removal procedure.
 - Lubricate the new O-ring seals with clean engine coolant.
7. Fill and bleed the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.

COOLANT PUMP

REMOVAL

CAUTION: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces, that enters the oil passages, coolant passages or the oil pan can cause engine failure.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Drain the cooling system.
3. Remove the engine front cover. For additional information, refer to **ENGINE - 3.5L** article.
4. Remove and discard the engine oil filter.
5. Rotate the crankshaft clockwise and align the timing marks on the variable camshaft timing (VCT) assemblies as shown in illustration.



A0021366

Fig. 6: Aligning Timing Marks On Variable Camshaft Timing (VCT) Assemblies
Courtesy of FORD MOTOR CO.

NOTE: The special tool will hold the camshafts in the top dead center (TDC) position.

6. Install the special tool onto the flats of the LH camshafts.

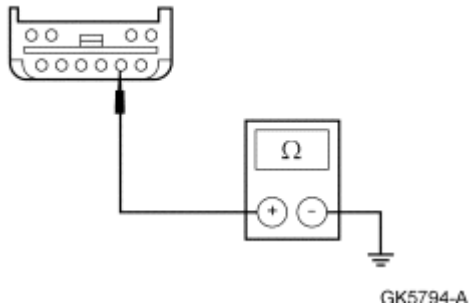


Fig. 7: Installing Special Tool Onto Flats Of LH Camshafts
Courtesy of FORD MOTOR CO.

NOTE: The special tool will hold the camshafts in the TDC position.

7. Install the special tool onto the flats of the RH camshafts.

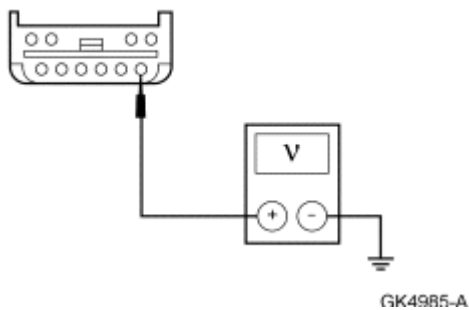


Fig. 8: Installing Special Tool Onto Flats Of RH Camshafts
Courtesy of FORD MOTOR CO.

8. Remove the 3 bolts and the RH VCT housing.

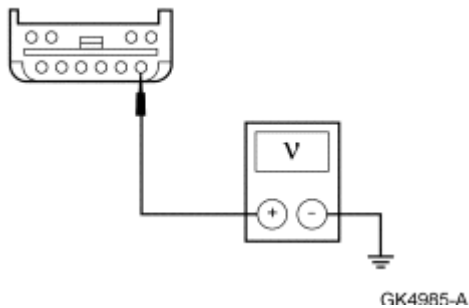


Fig. 9: Locating RH VCT Housing
Courtesy of FORD MOTOR CO.

9. Remove the 3 bolts and the LH VCT housing.

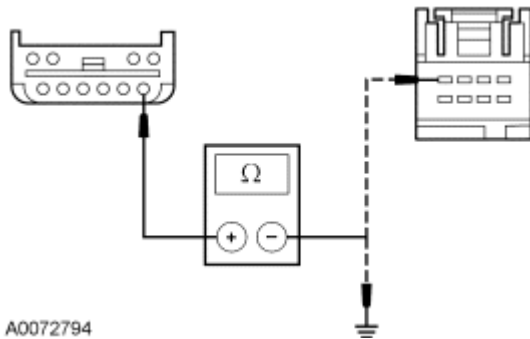


Fig. 10: Locating LH VCT Housing
Courtesy of FORD MOTOR CO.

10. Remove and discard the VCT housing seals.

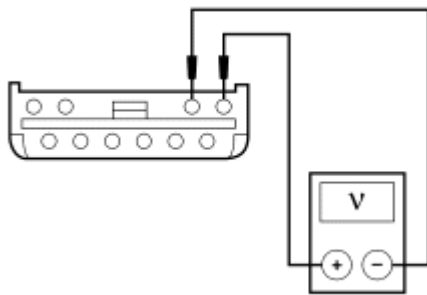


Fig. 11: Locating VCT Housing Seals
Courtesy of FORD MOTOR CO.

11. Remove the 2 bolts and the primary timing chain tensioner.

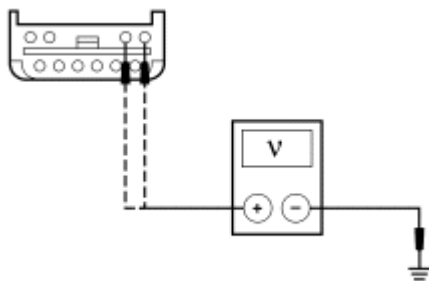
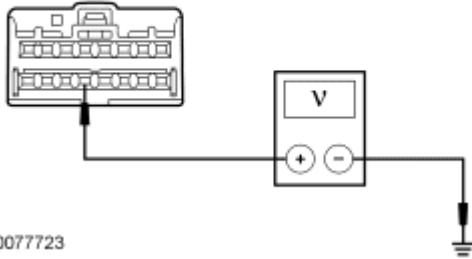


Fig. 12: Locating Primary Timing Chain Tensioner Bolts
Courtesy of FORD MOTOR CO.

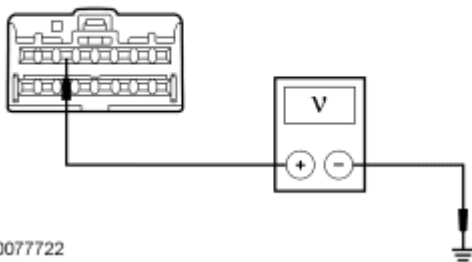
12. Remove the primary timing chain tensioner arm.



N0077723

Fig. 13: Locating Primary Timing Chain Tensioner Arm
Courtesy of FORD MOTOR CO.

13. Remove the 2 bolts and the lower LH primary timing chain guide.



N0077722

Fig. 14: Locating Lower LH Primary Timing Chain Guide Bolts
Courtesy of FORD MOTOR CO.

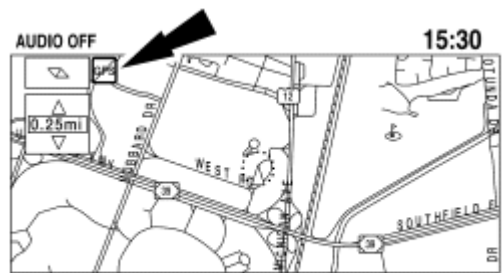
14. Remove the primary timing chain.



N0026638

Fig. 15: Locating Primary Timing Chain
Courtesy of FORD MOTOR CO.

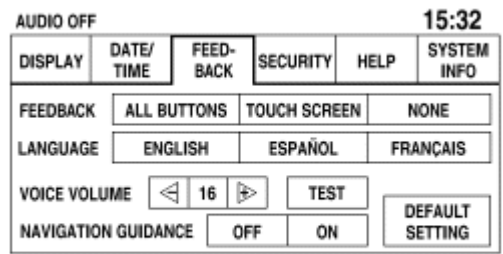
15. Remove the 2 bolts and the upper LH primary timing chain guide.



N0026638

Fig. 16: Locating Upper LH Primary Timing Chain Guide Bolts
Courtesy of FORD MOTOR CO.

16. Remove the RH primary timing chain guide lower bolt.



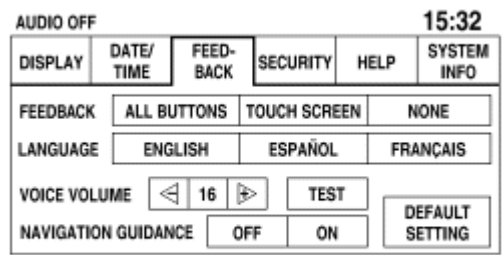
N0026637

Fig. 17: Locating RH Primary Timing Chain Guide Lower Bolt
Courtesy of FORD MOTOR CO.

NOTE: The RH primary timing chain guide must be repositioned to allow the coolant pump to be removed.

17. Loosen the RH primary timing chain guide upper bolt.

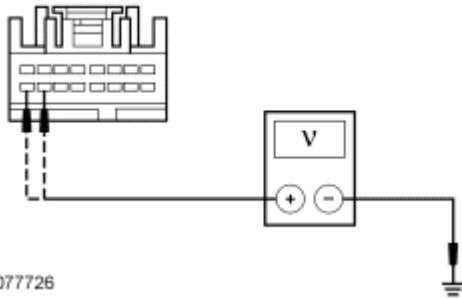
- Rotate the guide and tighten the bolt.



N0026637

Fig. 18: Locating RH Primary Timing Chain Guide Upper Bolt
Courtesy of FORD MOTOR CO.

18. Place clean lint-free shop towels in the oil pan opening to prevent coolant from entering the oil pan during coolant pump removal.
19. Remove the 8 bolts and the coolant pump.



N0077726

Fig. 19: Locating Coolant Pump Bolts
Courtesy of FORD MOTOR CO.

INSTALLATION

NOTE: Clean and inspect all sealing surfaces.

1. Install the coolant pump and the 8 bolts.
 - Tighten in the sequence shown in illustration to 10 N.m (89 lb-in).



N0026638

Fig. 20: Identifying Coolant Pump Bolts Tightening Sequence
Courtesy of FORD MOTOR CO.

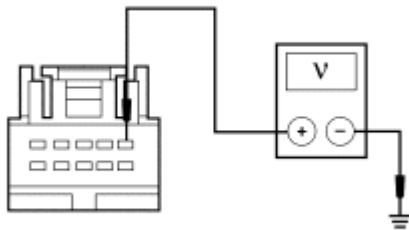
2. Remove all of the shop towels from the oil pan opening.

CAUTION: Any coolant that has accumulated in the oil pan must be drained from the pan and any residual coolant cleaned from the front of the engine and oil pan. Failure to remove all traces of the coolant can result in oil contamination and severe engine damage.

3. Remove the oil pan drain plug and allow any accumulated coolant to drain.
 - Remove any residual coolant from the front of the engine and the oil pan using regulated

compressed air and clean lint-free shop towels.

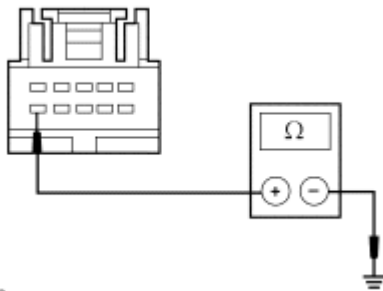
- Install the oil pan drain plug and tighten to 27 N.m (20 lb-ft).
4. Loosen the RH primary timing chain guide upper bolt.
- Position the RH primary timing chain guide and install the lower bolt.
 - Tighten the 2 bolts to 10 N.m (89 lb-in).



N0057331

Fig. 21: Locating RH Primary Timing Chain Guide Lower Bolt
Courtesy of FORD MOTOR CO.

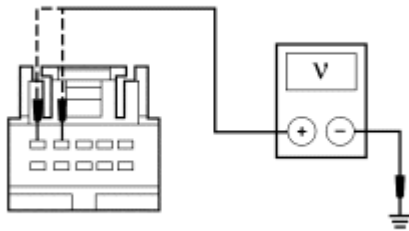
5. Install the primary timing chain with the colored links aligned with the timing marks on the VCT assemblies and the crankshaft sprocket.



N0057336

Fig. 22: Aligning Timing Marks On VCT Assemblies & Crankshaft Sprocket
Courtesy of FORD MOTOR CO.

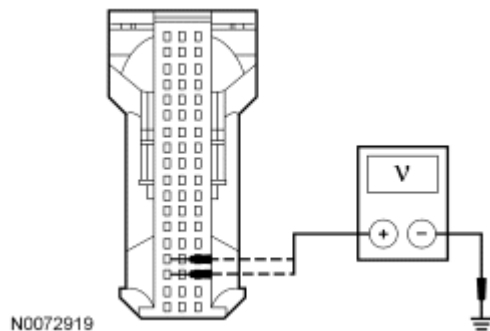
6. Install the upper LH primary timing chain guide and the 2 bolts.
- Tighten to 10 N.m (89 lb-in).



N0057337

Fig. 23: Locating Upper LH Primary Timing Chain Guide Bolts
Courtesy of FORD MOTOR CO.

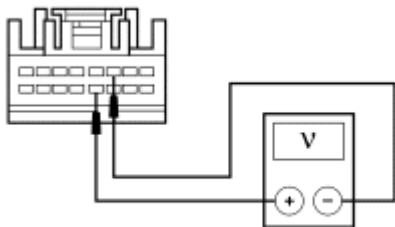
7. Install the lower LH primary timing chain guide and the 2 bolts.
 - Tighten to 10 N.m (89 lb-in).



N0072919

Fig. 24: Locating Lower LH Primary Timing Chain Guide Bolts
Courtesy of FORD MOTOR CO.

8. Install the primary timing chain tensioner arm.



N0077727

Fig. 25: Locating Primary Timing Chain Tensioner Arm
Courtesy of FORD MOTOR CO.

9. Reset the primary timing chain tensioner.
 - Rotate the lever counterclockwise.

- Using a soft-jawed vise, compress the plunger.
- Align the hole in the lever with the hole in the tensioner housing.
- Install a suitable lock pin.

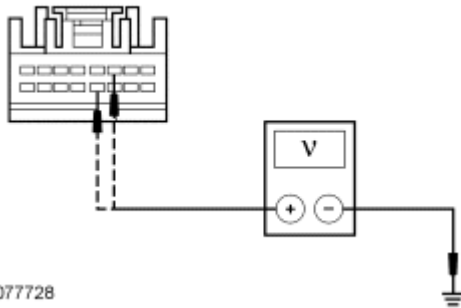


Fig. 26: Compressing Plunger Using A Soft-Jawed Vise
Courtesy of FORD MOTOR CO.

NOTE: It may be necessary to rotate the crankshaft slightly to remove slack from the timing chain and install the tensioner.

10. Install the primary tensioner and the 2 bolts.
 - Tighten to 10 N.m (89 lb-in).
 - Remove the lock pin.

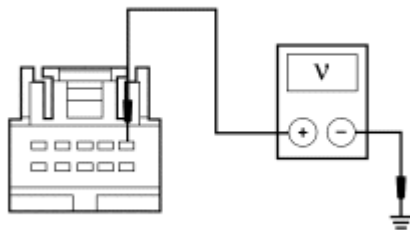
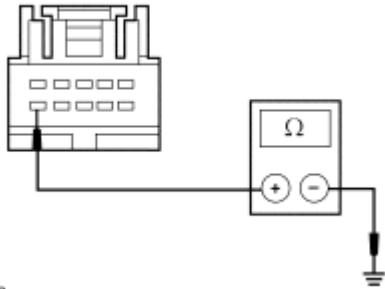


Fig. 27: Locating Primary Tensioner Bolts
Courtesy of FORD MOTOR CO.

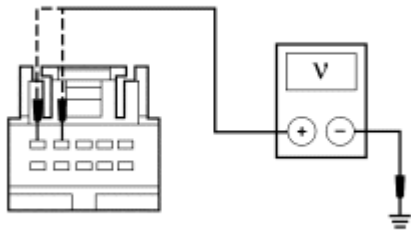
11. As a post check, verify correct alignment of all timing marks.



N0057336

Fig. 28: Verifying Correct Alignment Of All Timing Marks
Courtesy of FORD MOTOR CO.

12. Install new VCT housing seals.



N0057337

Fig. 29: Locating VCT Housing Seals
Courtesy of FORD MOTOR CO.

CAUTION: Make sure the dowels on the variable camshaft timing (VCT) housing are fully engaged in the cylinder head prior to tightening the bolts. Failure to follow this process will result in severe engine damage.

13. Install the LH VCT housing and the 3 bolts.
 - Tighten in the sequence shown in illustration to 10 N.m (89 lb-in).

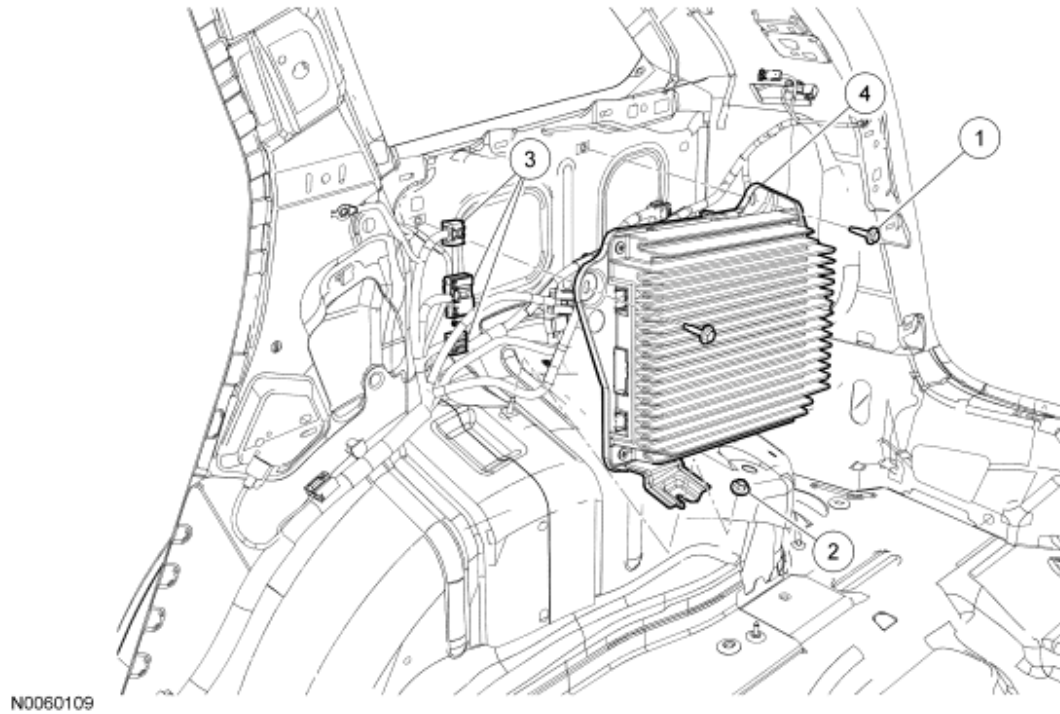
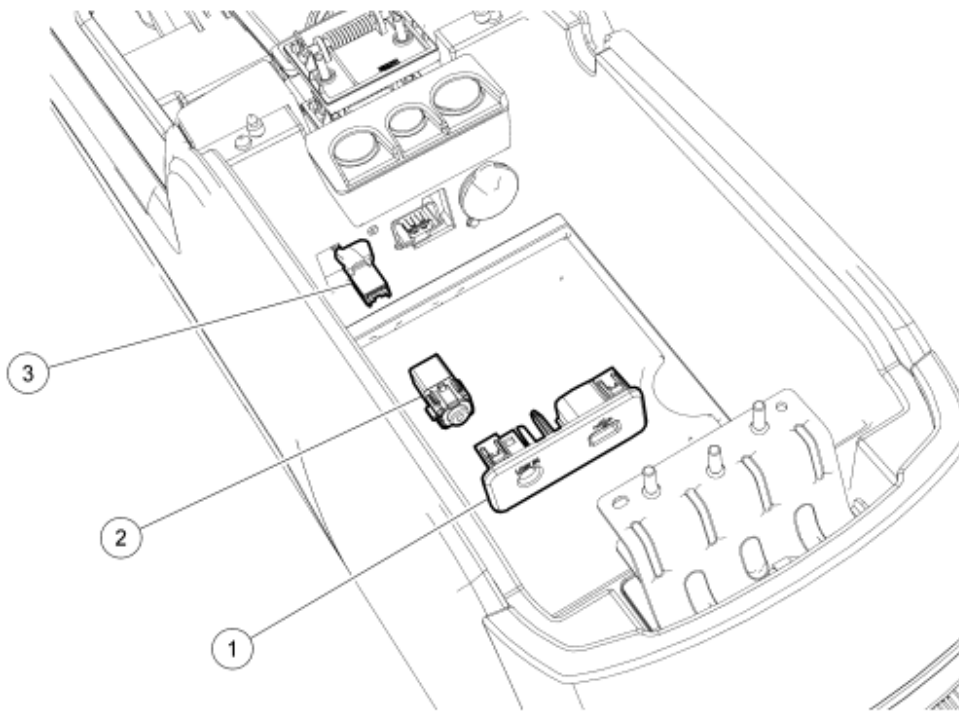


Fig. 30: Identifying LH VCT Housing Bolts Tightening Sequence
Courtesy of FORD MOTOR CO.

CAUTION: Make sure the dowels on the variable camshaft timing (VCT) housing are fully engaged in the cylinder head prior to tightening the bolts. Failure to follow this process will result in severe engine damage.

14. Install the RH VCT housing and the 3 bolts.
 - Tighten in the sequence shown in illustration to 10 N.m (89 lb-in).



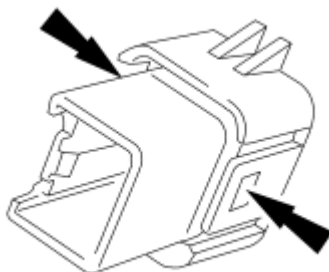
N0077732

Fig. 31: Identifying RH VCT Housing Bolts Tightening Sequence
Courtesy of FORD MOTOR CO.

NOTE: Lubricate the engine oil filter gasket with clean engine oil prior to installing the oil filter.

15. Install a new engine oil filter.
 - Tighten to 5 N.m (44 lb-in) and then rotate an additional 180 degrees.
16. Install the engine front cover. For additional information, refer to **ENGINE - 3.5L** article.
17. Fill and bleed the cooling system.

COOLING FAN MOTOR AND SHROUD



N0056277

Fig. 32: Exploded View Of Cooling Fan Motor & Shroud With Torque Specifications
Courtesy of FORD MOTOR CO.

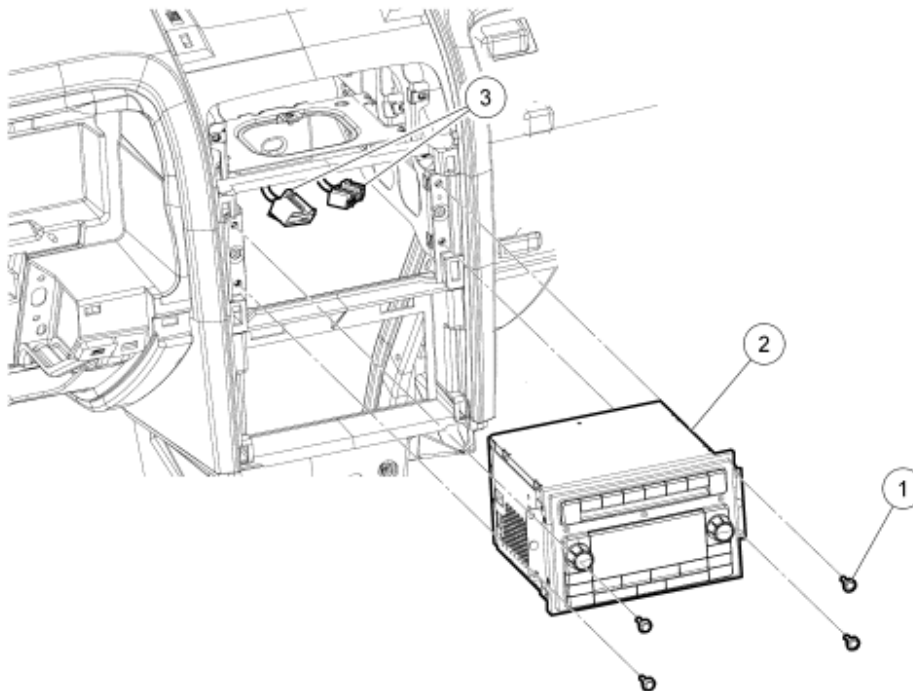
2008 Ford Edge SE

2008 ENGINE Engine Cooling - Edge & MKX

Item	Part Number	Description
1	13A506	Wiring harness retainer
2	8B274	Upper radiator hose
3	14A464	Cooling fan motor and shroud electrical connector
4	W503924	Cooling fan motor and shroud bolt (2 required)
5	8C607	Cooling fan motor and shroud
6	W503924	Oil cooler bracket bolt (if equipped)
7	-	Oil cooler bracket (if equipped)

REMOVAL AND INSTALLATION

1. Remove the air cleaner assembly. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
2. If equipped, detach the 2 block heater wiring clips from the radiator support.



N0060099

Fig. 33: Identifying Block Heater Wiring Clips
Courtesy of FORD MOTOR CO.

3. Detach the wiring harness retainers and position the harness aside.
4. Position aside the upper radiator hose from the cooling fan motor and shroud.
5. Disconnect the cooling fan motor and shroud electrical connector.
6. If equipped, remove the bolt and position aside the oil cooler bracket.

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- To install, tighten to 6 N.m (53 lb-in).
- 7. Remove the 2 bolts and the cooling fan motor and shroud.
 - To install, tighten to 6 N.m (53 lb-in).
- 8. To install, reverse the removal procedure.

RADIATOR

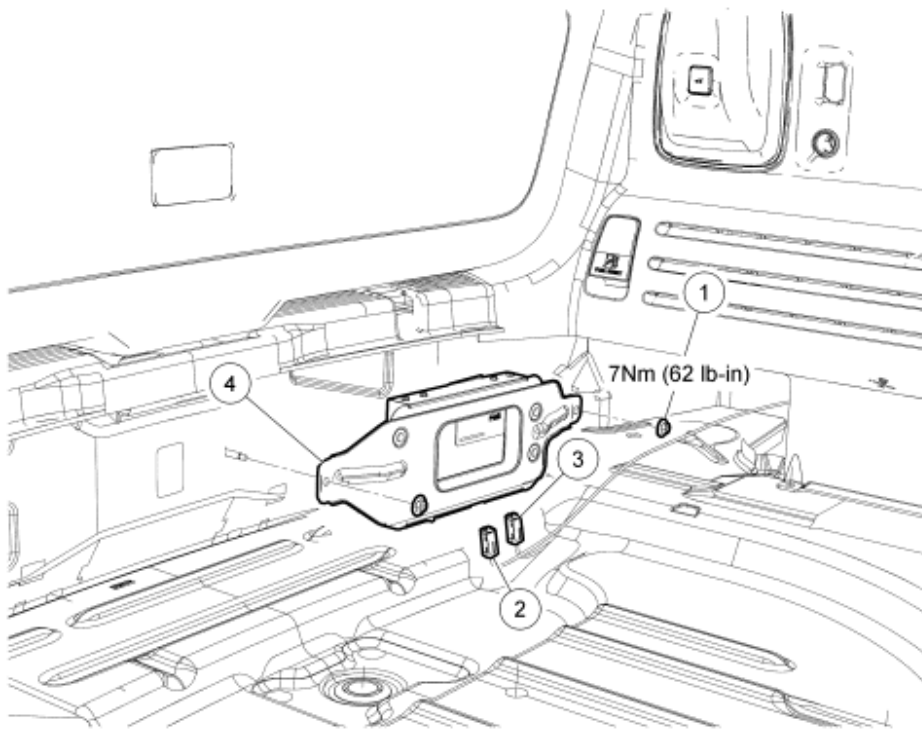
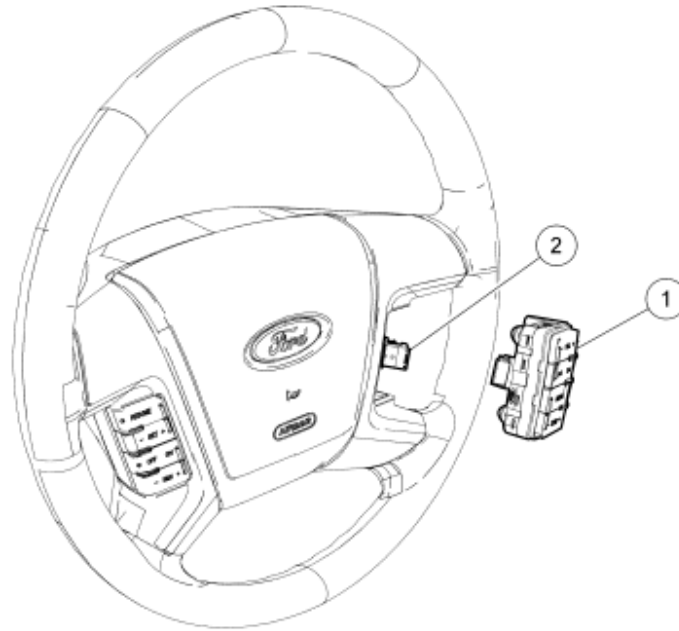


Fig. 34: Exploded View Of Radiator (Without A/C)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	8B274	Upper radiator hose
2	8C633	Lower degas bottle hose
3	8B273	Lower radiator hose
4	8A194	Upper radiator support bracket (2 required)

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2008 ENGINE Engine Cooling - Edge & MKX



N0035298

Fig. 35: Exploded View Of Radiator (With A/C) With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
5	-	A/C condenser bolt (2 required)
6	19710	A/C condenser
7	8005	Radiator
8	19D738	Pin-type retainer (2 required)
9	W707400	Compressor-to-condenser discharge line bracket bolt

REMOVAL AND INSTALLATION

1. Drain the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.
2. Remove the cooling fan motor and shroud. For additional information, refer to **COOLING FAN MOTOR AND SHROUD**.
3. Remove the front bumper cover. For additional information, refer to **BUMPERS** article.
4. Disconnect the upper radiator hose and lower degas bottle hose from the radiator.
5. Disconnect the lower radiator hose from the radiator.
6. Remove the compressor-to-condenser discharge line bracket bolt.
 - To install, tighten to 7 N.m (62 lb-in).
7. Lift and remove the tabs from the radiator support and position the radiator towards the engine.

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8. Detach the 2 pin-type retainers from the radiator.
9. Remove the 2 A/C condenser bolts from the radiator and separate the condenser from the radiator.
 - To install, tighten to 10 N.m (89 lb-in).
10. Remove the radiator.
11. To install, reverse the removal procedure.
12. Fill and bleed the cooling system. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING**.

DEGAS BOTTLE

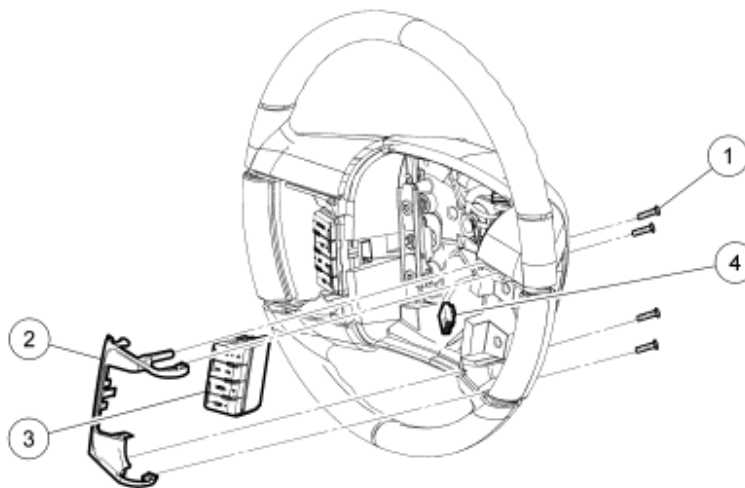


Fig. 36: Exploded View Of Degas Bottle With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	8W005	Upper degas bottle hose
2	8C633	Lower degas bottle hose
3	N807658	Degas bottle-to-fender bolt (2 required)
4	N807658	Degas bottle-to-washer bottle screw
5	8A080	Degas bottle
6	8100	Pressure relief cap

REMOVAL AND INSTALLATION

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

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1. Release the pressure in the cooling system by slowly turning the pressure relief cap one half turn counterclockwise. When the pressure is released, remove the pressure relief cap.
2. Using hose clamp pliers, clamp the lower degas bottle hose.
3. Using a suitable suction device, siphon the coolant from the degas bottle.
4. Disconnect the upper degas bottle hose and position it aside.
5. Disconnect the lower degas bottle hose and position it aside.
6. Remove the degas bottle-to-washer bottle screw.
7. Remove the 2 bolts and the degas bottle.
 - To install, tighten to 9 N.m (80 lb-in).
8. To install, reverse the removal procedure.
9. Fill the degas bottle. For additional information, refer to **COOLING SYSTEM DRAINING, FILLING AND BLEEDING** for the recommended coolant mixture and fill level.